

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1101

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 60%

QUESTIONS: 6

Total Marks:60

Annexure A

Application Based Questions

Code of Conduct:

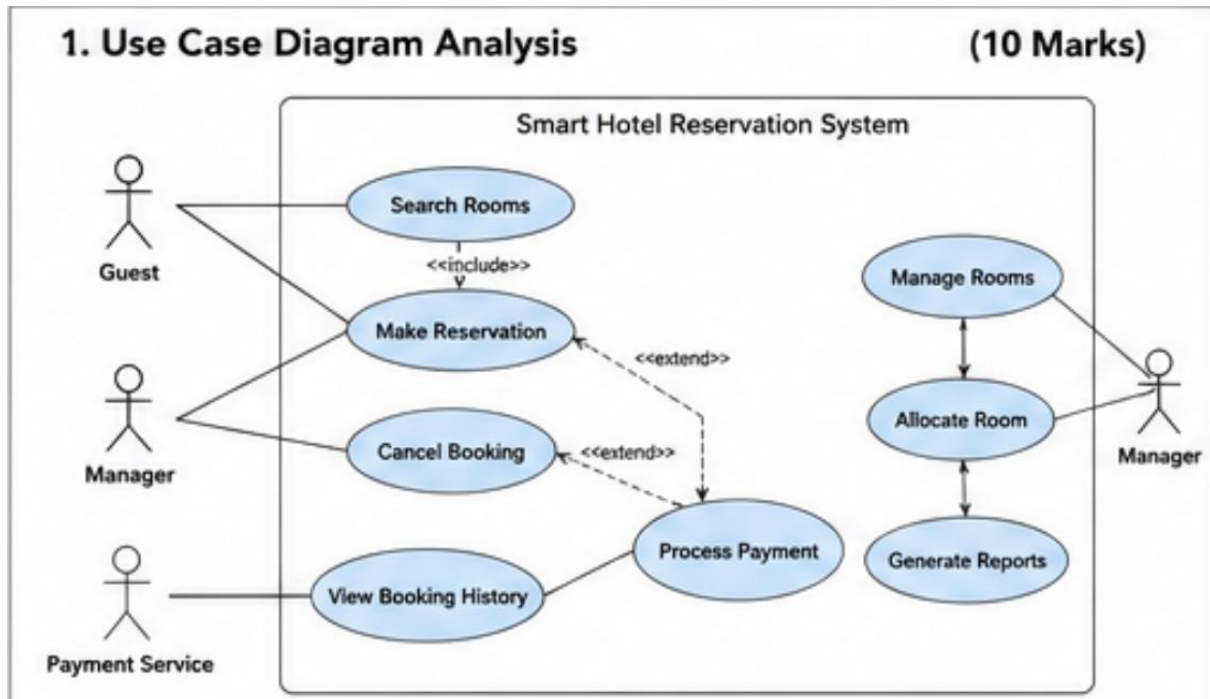
*I Mathusri P-192311363 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Mathusri P

Annexure A

Application Based Questions

Question 1: Use Case Diagram Analysis – Smart Hotel Reservation System



A Smart Hotel Reservation System allows guests to search for available rooms, reserve rooms, cancel bookings, make payments, and receive booking confirmation. The system also allows hotel managers to allocate rooms, update room availability, and generate reports. An external payment service processes financial transactions.

Actors Identified

- Guest
- Registered Guest
- Hotel Manager
- Payment Service
- Notification Service

The Registered Guest can be shown as a specialized actor of Guest because a registered guest can perform all basic guest activities and also access booking history or saved details.

Main Use Cases

- Search Room
- View Room Details
- Make Reservation
- Cancel Reservation
- Process Payment
- Receive Confirmation
- Allocate Room
- Update Room Availability
- Generate Reports
- Send Notification

Relationships

- Make Reservation **includes** Search Room.
- Make Reservation **includes** Check Availability.
- Make Reservation **includes** Process Payment.
- Process Payment **includes** Validate Transaction.
- Receive Confirmation **extends** Make Reservation.
- Cancel Reservation **includes** Verify Booking.
- Refund Payment **extends** Cancel Reservation.
- Registered Guest is a specialization of Guest.

Analysis

The optimized diagram clearly separates the responsibilities of guests, managers, and external services. The include relationship is used for compulsory functions, while the extend relationship is used for optional or conditional functions. Generalization helps represent common and specialized actor behavior.

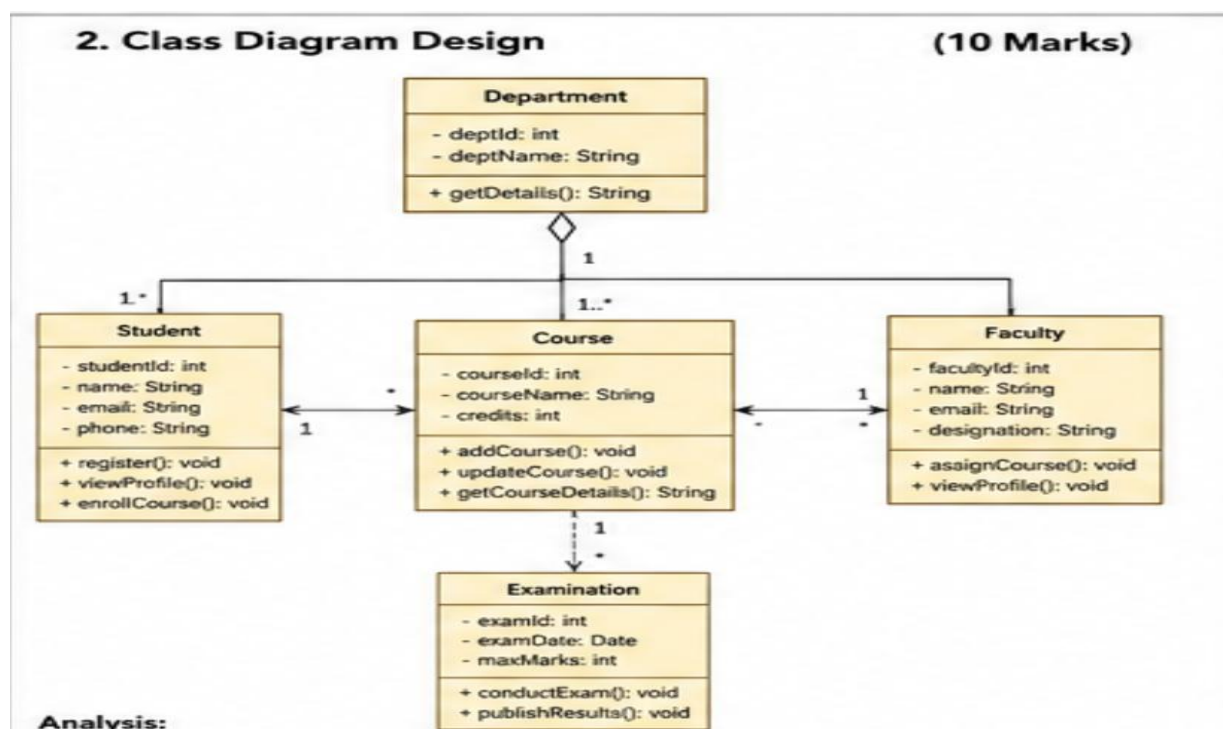
Benefits

- Improves understanding of actor responsibilities.
- Prevents duplication of use cases.
- Clearly represents optional and compulsory functions.
- Supports future system expansion.
- Improves communication between developers and stakeholders.

Conclusion

The Use Case Diagram provides a clear functional view of the Smart Hotel Reservation System and ensures that all users and external services interact with the correct system functions.

Question 2: Class Diagram Design – University Management System



A University Management System manages students, faculty members, courses, departments, and examinations. A well-designed Class Diagram must include classes, attributes, methods, and relationships.

Main Classes and Attributes

Student

Attributes:

- studentId
- studentName
- email
- phone
- semester

Methods:

- registerCourse()
- viewResult()
- payFees()
- updateProfile()

Faculty

Attributes:

- facultyId
- facultyName
- designation
- email
- specialization

Methods:

- teachCourse()
- markAttendance()
- enterMarks()
- viewSchedule()

Course

Attributes:

- courseId
- courseName
- credits
- duration

Methods:

- addStudent()
- removeStudent()
- assignFaculty()
- updateCourse()

Department

Attributes:

- departmentId
- departmentName
- location

Methods:

- addCourse()
- assignFaculty()
- manageStudents()
- generateReport()

Examination

Attributes:

- examId
- examDate
- examType
- maximumMarks

Methods:

- scheduleExam()
- publishResult()
- updateMarks()
- generateResult()

Relationships

- A Department aggregates many Students.
- A Department aggregates many Faculty members.
- A Department offers many Courses.
- A Student enrolls in many Courses.
- A Faculty member teaches one or more Courses.
- A Course may have many Examinations.
- Student and Faculty can inherit common attributes from a Person class.

Analysis

Association represents relationships such as Student enrolling in Course. Aggregation represents Department containing Students, Faculty, and Courses while these entities can exist independently. Inheritance avoids duplication by placing common details such as name, email, and phone in a Person superclass.

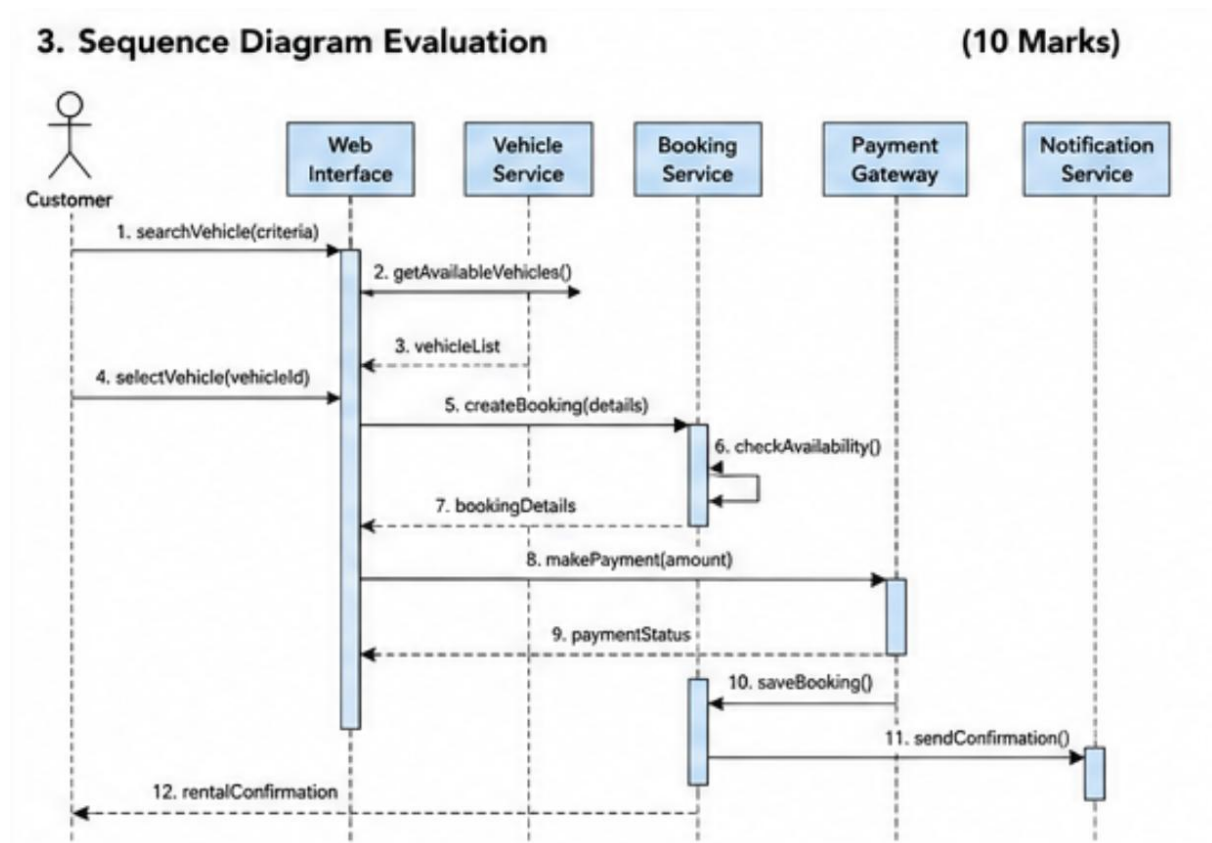
Benefits

- Improves modular design.
- Reduces duplicate attributes.
- Clearly defines class responsibilities.
- Supports reuse through inheritance.
- Makes future maintenance easier.

Conclusion

The Class Diagram provides a structured representation of the University Management System and clearly shows the static relationships among all major classes.

Question 3: Sequence Diagram Evaluation – Vehicle Rental System



A Vehicle Rental System allows customers to search for a vehicle, select it, confirm booking details, make payment, and receive rental confirmation. A Sequence Diagram shows how objects interact in chronological order.

Objects Identified

- Customer
- Vehicle Rental Interface
- Vehicle Search Service
- Vehicle Database
- Booking Service
- Payment Gateway
- Notification Service

Message Sequence

1. The Customer enters vehicle search details.
2. The interface sends the search request to the Vehicle Search Service.
3. The Vehicle Search Service requests available vehicles from the Vehicle Database.
4. The database returns matching vehicle details.
5. The interface displays the available vehicles.
6. The Customer selects a vehicle.
7. The interface sends booking details to the Booking Service.
8. The Booking Service verifies vehicle availability.
9. The Customer confirms the booking.
10. The Booking Service sends payment details to the Payment Gateway.
11. The Payment Gateway processes the payment.

12. The payment status is returned to the Booking Service.
13. The Booking Service creates the rental booking.
14. The Notification Service sends confirmation to the Customer.

Alternative Conditions

- When no vehicle is available, the system displays an unavailable message.
- When payment fails, the booking is not confirmed.
- When payment succeeds, the system generates a booking ID and confirmation.

Analysis

The Sequence Diagram clearly shows the order of communication among the customer, user interface, services, database, and payment gateway. It also identifies missing system components and ensures that booking is confirmed only after successful payment.

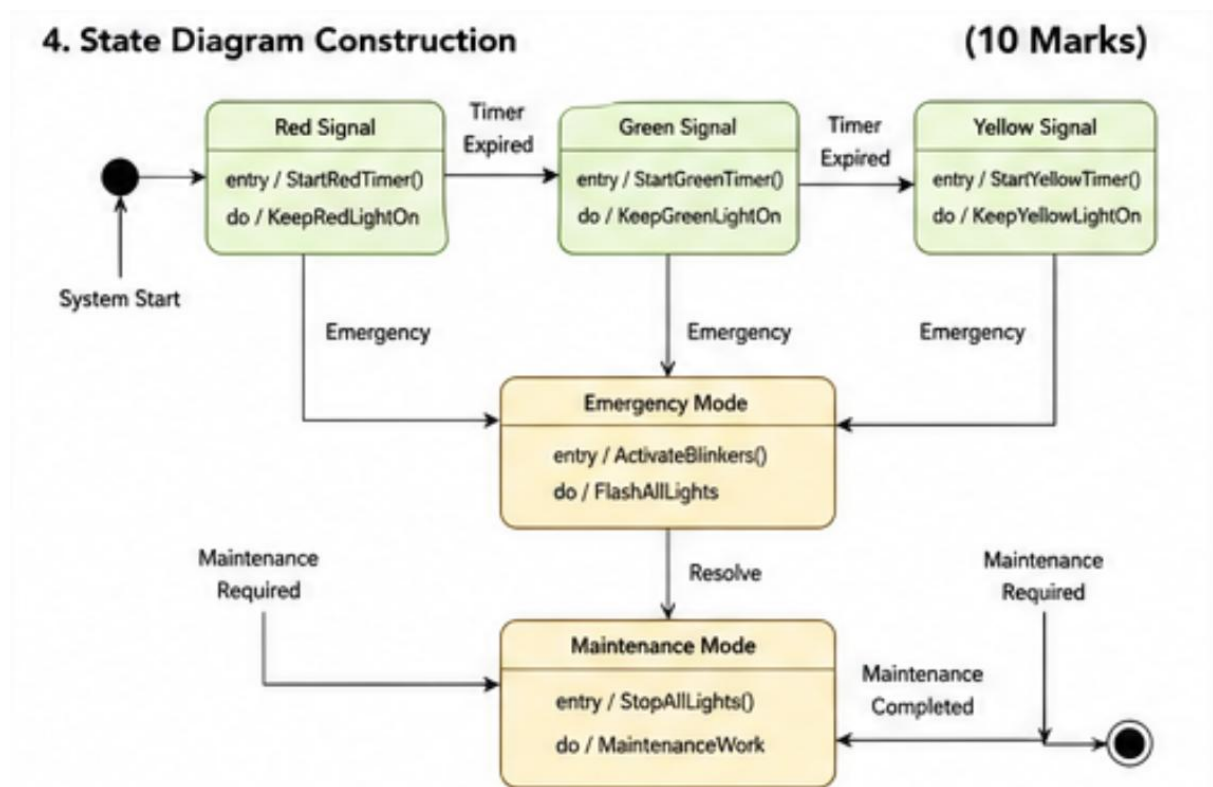
Benefits

- Shows time-ordered interactions.
- Helps identify missing messages.
- Clarifies service responsibilities.
- Supports testing and debugging.
- Improves system documentation.

Conclusion

The Sequence Diagram provides a complete representation of the vehicle booking process and ensures proper communication between all system components.

Question 4: State Diagram Construction – Traffic Signal Control System



A Traffic Signal Control System changes its behavior based on different operating states. The major states are Red Signal, Green Signal, Yellow Signal, Emergency Mode, and Maintenance Mode.

Main States

Red Signal

Vehicles must stop. After the red timer ends, the system changes to Green Signal.

Green Signal

Vehicles are allowed to move. After the green timer ends, the system changes to Yellow Signal.

Yellow Signal

Vehicles are warned that the signal will soon turn red. After the yellow timer ends, the system changes to Red Signal.

Emergency Mode

The normal signal cycle is interrupted when an ambulance, fire service, or police vehicle is detected. The required road is given priority.

Maintenance Mode

The normal signal operation is stopped when maintenance is required. The signal may use blinking yellow lights until maintenance is completed.

Transitions

- Initial State to Red Signal when the system starts.
- Red Signal to Green Signal when the red timer expires.
- Green Signal to Yellow Signal when the green timer expires.
- Yellow Signal to Red Signal when the yellow timer expires.
- Any normal state to Emergency Mode when an emergency vehicle is detected.
- Emergency Mode to Red Signal when the emergency condition ends.
- Any state to Maintenance Mode when a fault is detected.
- Maintenance Mode to Red Signal after repair is completed.
- Any state to Final State when the system is shut down.

Analysis

The State Diagram ensures that every signal change is controlled by a valid event or condition. Emergency and maintenance states are given higher priority than the normal signal cycle.

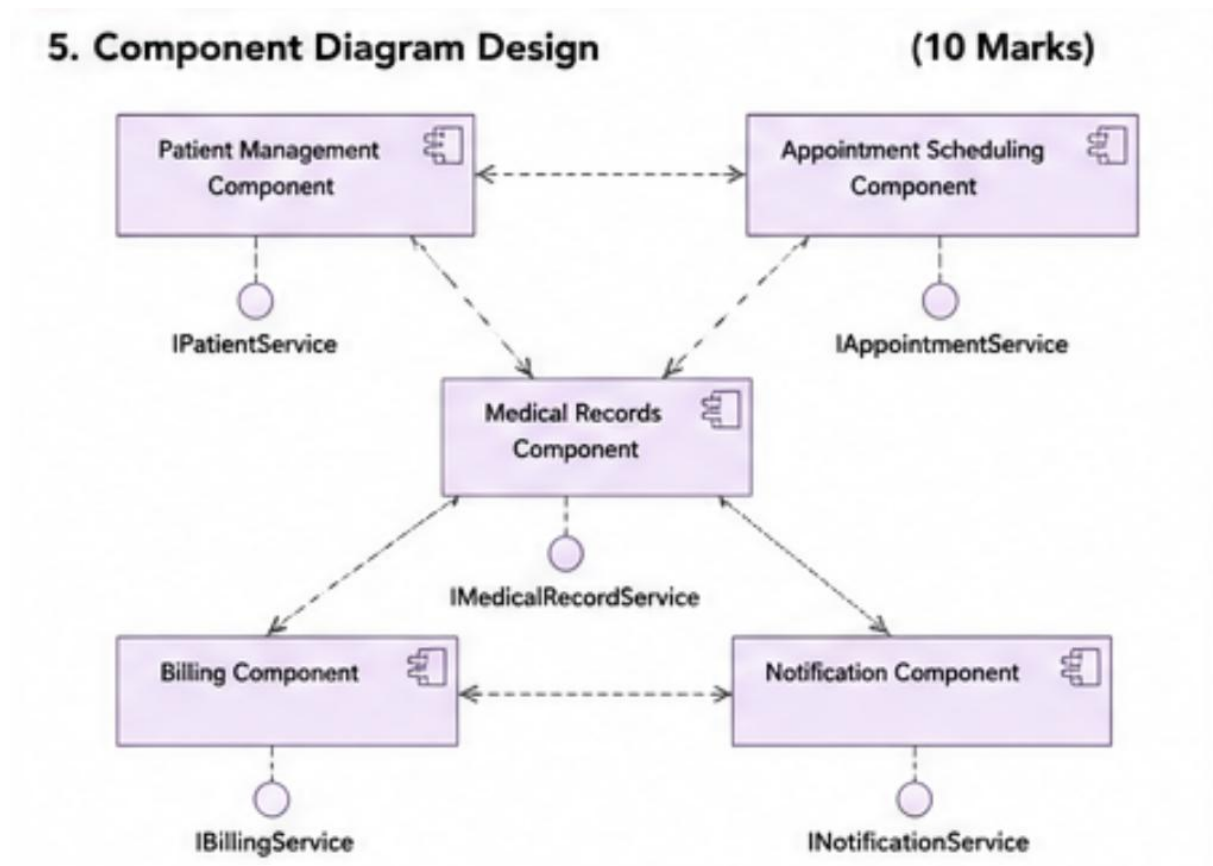
Benefits

- Clearly represents system behavior.
- Prevents invalid signal changes.
- Supports emergency handling.
- Improves road safety.
- Helps developers test all possible states.

Conclusion

The State Transition Diagram provides an organized view of traffic signal behavior and ensures safe and reliable transitions under normal, emergency, and maintenance conditions.

Question 5: Component Diagram Design – Smart Healthcare Portal



A Smart Healthcare Portal contains several software components that work together to manage patients, appointments, medical records, billing, and notifications.

Major Components

User Interface

Provides separate interfaces for patients, doctors, and administrators.

Authentication Service

Verifies users and controls access based on roles.

Patient Management Component

Manages patient registration, personal details, and profile updates.

Appointment Scheduling Component

Manages doctor availability, appointment booking, rescheduling, and cancellation.

Medical Records Component

Stores diagnosis, treatment history, prescriptions, and laboratory reports.

Billing Component

Calculates consultation charges, treatment costs, and payment status.

Notification Service

Sends appointment reminders, payment confirmations, and medical alerts.

Database Component

Stores patient, doctor, appointment, billing, and medical information.

External Payment Gateway

Processes online payments securely.

Dependencies

- User Interface depends on Authentication Service.
- Patient Management depends on the Database.
- Appointment Scheduling depends on Patient Management and Doctor Schedule.
- Medical Records depends on Patient Management and the Database.
- Billing depends on Appointment Scheduling and Medical Records.
- Billing connects to the External Payment Gateway.
- Notification Service depends on Appointment Scheduling and Billing.

Interfaces

- Authentication Interface
- Patient Data Interface
- Appointment Interface
- Medical Record Interface
- Billing Interface
- Notification Interface
- Payment Interface

Analysis

The Component Diagram separates the healthcare portal into independent modules. Each component communicates through a defined interface. This reduces coupling and allows individual components to be changed or upgraded without affecting the entire application.

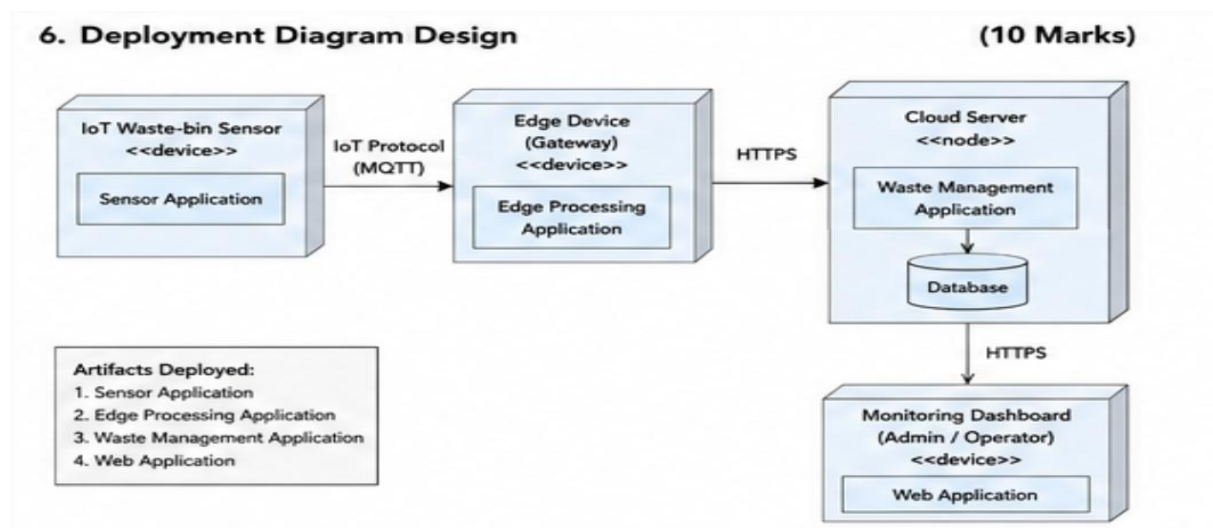
Benefits

- Supports modular development.
- Improves scalability.
- Simplifies maintenance.
- Protects sensitive medical data.
- Makes testing easier.

Conclusion

The Component Diagram provides a clear architectural view of the Smart Healthcare Portal and shows how all modules, services, and external systems interact.

Question 6: Deployment Diagram Design – Smart City Waste Management System



A Smart City Waste Management System uses IoT waste-bin sensors, edge devices, cloud servers, databases, and monitoring dashboards to track bin levels and optimize waste collection.

Deployment Nodes

Smart Waste-Bin Sensor Node

Contains sensors that measure:

- Waste level
- Weight
- Temperature
- Gas level
- Battery status

Artifact:

- Sensor Monitoring Firmware

Edge Gateway

Collects data from nearby waste bins, filters it, and sends it to the cloud.

Artifacts:

- Data Collection Service
- Data Preprocessing Module

Communication Network

Provides communication through Wi-Fi, LoRaWAN, 4G, or 5G.

Cloud Application Server

Processes sensor data, identifies full bins, and generates collection schedules.

Artifacts:

- Waste Management Application
- Route Optimization Service
- Alert Service

Cloud Database Server

Stores bin information, sensor readings, alerts, collection records, and vehicle routes.

Artifact:

- Waste Management Database

Monitoring Dashboard

Used by municipal officers to monitor bins, view alerts, and assign collection vehicles.

Artifact:

- Web or Mobile Dashboard

Notification Server

Sends alerts to collection staff when bins become full or sensors fail.

Artifact:

- Notification Service

Communication Links

- Sensors communicate with the Edge Gateway using wireless communication.
- Edge Gateway sends processed data to the Cloud Server using the internet.
- Cloud Server stores information in the Database Server.
- Dashboard communicates with the Cloud Server through secure HTTPS.
- Notification Server sends alerts to collection staff.
- Collection vehicles may send GPS data to the Cloud Server.

Analysis

The Deployment Diagram shows both physical devices and deployed software artifacts. Edge processing reduces network traffic, while cloud infrastructure provides large-scale storage and analytics. Secure communication ensures that sensor and location data are protected.

Benefits

- Enables real-time waste monitoring.
- Reduces unnecessary collection trips.
- Optimizes vehicle routes.
- Prevents overflowing waste bins.
- Improves city cleanliness.
- Supports scalable deployment.

Conclusion

The Deployment Diagram clearly represents the physical architecture of the Smart City Waste Management System and explains how IoT devices, edge gateways, cloud services, databases, and dashboards communicate.

RUBRICS FOR CALCULATION

Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand